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FACULTY OF COMPUTING AND MATHEMATICS
DEPARTMENT OF INFORMATION TECHNOLOGY
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
MODULE NAME: INNOVATION AND ENTREPRENEURSHIP

MODULE CODE: ITU08216

YEAR THREE

ACADEMIC YEAR: 2025/2026

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ASSIGNMENT TITLE: Innovation Opportunities and Entrepreneurial Solutions to Socio-Economic Challenges in Tanzania

SECTOR IN TANZANIA: Healthcare at kigamboni.

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1.1. TASK 1. PROBLEM IDENTIFICATION

1.2.The background of the healthcare sector

Kigamboni Hospital is a public district hospital in Somangila, Dar es Salaam, which officially opened on January 6, 2020. It provides OPD, IMCI, diabetes care, mental health services, and minor surgeries. The hospital uses the Health Facility Registry System (HFRS), Hospital Information System (HIS), Laboratory Information System (LIS), Pharmacy Management System, and modern diagnostic equipment including CT scanners and dialysis machines. These machines help doctors and nurses take care of people better.

1.3.The description of the problem

The big problem we found is that some healthcare workers do not know how to use machines. When we talked to people, they said they did not get practice using new medical equipment.

1.4.The magnitude, of the problem

This problem hurts healthcare workers, patients and healthcare facilities. The bad things that happen because of this problem are,

- a) It takes longer to treat patients.
- b) medical machines are used wrongly.
- c) Expensive equipment gets damaged.
- d) The healthcare people get is not very good.
- e) It costs more to fix equipment.

1.5.Evidence supporting the problem

Our group talked to healthcare workers in Kigamboni. What we found out was that

- a) A lot of the people we talked to said they need training to do their jobs well.
- b) Some workers had a time using new medical devices.
- c) Most of the people we talked to thought it was an idea to have simple digital training materials that they could use.

2.1. TASK 2. FIELD DATA COLLECTION

2.2.Data collection method

Primary data was collected exclusively through structured face-to-face interviews. This allowed us to gather in-depth, qualitative, and quantitative feedback directly from healthcare workers regarding their experiences with new medical equipment.

2.3.Sample Size

To ensure a reliable representation of the challenge, we interviewed a total of 30 respondents from various departments within Kigamboni Hospital.

2.4. Respondent characteristics

The people we talked to were 30 healthcare workers who worked at Kigamboni Hospital. They had amounts of work experience. Roles: 15 Nurses, 5 Medical Doctors, 5 Laboratory Technicians, and 5 Medical Equipment Operators.

2.5. Ethical considerations

When we were collecting information,

- a) We told the people we talked to why we were doing the study.
- b) They did not have to participate if they did not want to.
- c) We kept their information private.
- d) We were honest and respectful during the interviews.

3.1. TASK 3. ANALYSIS OF INNOVATION AND ENTREPRENEURSHIP CONCEPTS

3.2. Innovation in the healthcare sector

When we look at facilities like Kigamboni Hospital, we see a lot of modern diagnostic machines and digital systems being brought in to improve care. However, true innovation isn't just about placing new machines in a room; it is about ensuring the staff can comfortably and safely interact with that technology. Right now, the technology is advancing faster than the training methods. Innovation in this sector needs to focus on the "human element" finding smarter and faster ways to upskill our healthcare workers.

3.3. Entrepreneurship opportunities

As IT students, we realize that this training gap isn't just a hospital problem; it is a clear business opportunity. The struggle to understand new equipment opens the door for several entrepreneurial ventures, such as:

- **Health-EdTech Startups:** Building mobile learning platforms tailored specifically for African healthcare workers.
- **Digital Content Creation:** A business focused on producing high-quality, short Kiswahili video tutorials and licensing them to the Ministry of Health or private hospitals.
- **Consulting and Support Services:** Offering tech-adoption services to help hospitals transition smoothly when they buy complex digital equipment.

3.4. Importance of innovation

Innovation is important because it

- **Protects Patients:** It directly reduces medical errors caused by guessing how a machine works.
- **Saves Time:** Nurses and doctors can get back to treating people instead of spending a lot of minutes figuring out machine settings.
- **Saves Money:** It prevents expensive hospital equipment from getting damaged due to user error.
- **Builds Confidence:** It empowers healthcare workers by giving them support right in their pockets.

3.5. Tanzanian example

To prove that digital health solutions work in our local context, we can look at several practical innovations already succeeding in Tanzania:

- **Digital Hospital Systems:** The government has successfully rolled out systems like the Health Facility Registry System (HFRS) and Hospital Information System (HIS), which are already actively used at Kigamboni Hospital. This shows that our hospitals are ready to embrace digital workflows.
- **Zipline Drones:** Operating in regions like Dodoma and Mwanza, this company uses autonomous drones to deliver emergency blood and medical supplies, showing how high-tech innovation solves local logistics problems.
- **Jamii Africa:** A local startup that uses basic mobile phone technology (USSD) to provide affordable micro-health insurance to the informal sector.
- **Elsa Health:** A Tanzanian health-tech tool that uses artificial intelligence to assist local healthcare workers with diagnostic decisions.

4.1. TASK 4. CREATIVITY AND IDEA GENERATION

4.2. Using SCAMPER

- a) Substitute. Instead of using printed manuals we can use short video tutorials.
- b) Combine. We can combine QR codes with mobile learning videos.
- c) Adapt. We can change the training materials to be in Kiswahili.
- d) Modify. We can make the training videos short and easy to understand.
- e) Put To Another Use. We can use smartphones to train healthcare workers during their time.
- f) Eliminate. We can reduce the need for classroom training.
- g) Reverse. We can let healthcare workers learn before they use the equipment.

4.3. Design thinking process

- a) Empathize. We talk to healthcare workers to understand their challenges.
- b) Define. Healthcare workers lack skills in using medical devices.
- c) Ideate. We generate ideas like printed manuals, workshops and mobile learning.

- d) Prototype. We create a mobile training system with QR codes.
- e) Test. We let healthcare workers use the system and collect feedback to make it better.

4.4. Alternative solutions

- a) workshops.
- b) Printed user manuals.
- c) Hiring trainers.
- d) Online training videos.

4.5. Final selected solution

Our group chose a Smart QR Training System.

Every medical device will have a QR code. When scanned it opens a Kiswahili video explaining how to use the device. This solution is simple, affordable and easy to use.

5.1. TASK 5. OPPORTUNITY IDENTIFICATION AND EVALUATION

5.2. SWOT analysis

Strengths

- a) Easy to use.
- b) Low cost.
- c) Supports learning.
- d) Reduces equipment misuse.

Weaknesses

- a) Requires smartphones.
- b) Internet may be limited in some areas.

Opportunities

- a) Can be used in hospitals.
- b) Can be expanded across Tanzania.
- c) Supports healthcare improvement.

Threats

- a) People may resist change.
- b) Limited funding.
- c) Poor internet connection.

5.3. Opportunity evaluation

- a) Market demand

Hospitals need affordable training solutions because new medical devices are always being introduced.

- b) Technical feasibility

The innovation only requires QR codes, smartphones and short instructional videos.

- c) Financial feasibility

The project requires investment because QR codes are inexpensive and videos can be produced once and used many times.

- d) Competitive advantage

The solution is simple, available in Kiswahili. Allows healthcare workers to learn anytime.

- e) Scalability

The innovation can easily be expanded to hospitals and health centers across Tanzania.

6.1. TASK 6. INNOVATION MANAGEMENT PLAN

6.2. Idea generation

The idea came naturally while we were talking to the healthcare workers at Kigamboni Hospital. They shared that while having new medical devices is great, struggling to figure out how to use them is incredibly stressful. We brainstormed ways to deliver training right at the machine without getting in their way, which led us to the idea of smartphone-accessible videos

6.3. Screening

We looked at a few different options to solve this, like printing out translated manuals or organizing weekly training workshops. However, we realized manuals get lost or torn easily, and workshops take nurses away from their patients. We chose the Smart QR Training System because it is affordable, simple, and uses the phones the staff already have with them.

6.4. Prototype development

To test this out without overwhelming anyone, we will start small. We plan to pick just three common machines in one department. We will record short, clear, 2-minute instruction videos in Kiswahili for each machine. Then, we will print QR code stickers, place them on the machines, and ask a few workers to try scanning and learning for a week to see if it actually helps their daily routine.

6.5. Commercialization

To turn this into a working business, we would sell the system as a service directly to hospital management. A hospital would pay a reasonable yearly subscription fee. In return, they get our durable QR stickers and access to our library of training videos. Whenever the hospital buys a new type of machine, we would film a new tutorial for them as part of the service.

6.6. Diffusion

We will roll this out step by step. First, we will focus entirely on proving that the system works well at Kigamboni Hospital. Once we can show that it saves time and reduces mistakes there, we will share those positive results with other district hospitals in Dar es Salaam to get them on board. Eventually, we hope to partner with the Ministry of Health to make this a standard training tool in hospitals across Tanzania.

7.1.CONCLUSION

The study found that healthcare workers in Kigamboni lack skills in using devices. This affects the quality of healthcare services and patient safety.

The proposed Smart QR Training System provides a affordable and innovative solution. It will improve workers skills reduce mistakes and support healthcare services. The Smart QR Training System is a solution for healthcare workers in Kigamboni. Healthcare workers in Kigamboni will be able to use devices correctly with the help of the Smart QR Training System. The Smart QR Training System will make healthcare services better, in Kigamboni.

8.1. APPENDICES

APPENDIX A: DATA COLLECTION INSTRUMENT (INTERVIEW GUIDE)

Target Audience: Healthcare Workers at Kigamboni Hospital

Interviewer Instructions: Ask the following questions sequentially. Allow the respondent to elaborate on open-ended questions. Record responses accurately.

Part 1: Respondent Demographics & Professional Profile

1. Could you please state your primary profession or role here at the hospital?
2. How many years of experience do you have working in the healthcare sector?
3. Which specific ward or department do you primarily work in?

Part 2: Equipment Usage & Current Training Challenges

4. On a scale of 1 to 5 (with 1 being not confident at all, and 5 being highly confident), how confident do you feel operating newly procured digital medical equipment?
5. Have you ever faced difficulties operating a medical device specifically due to a lack of formal training?
6. (If Yes to Q5) Could you describe the direct consequences of those difficulties?
7. How do you currently go about learning to use a new machine or device in your department?
8. Overall, how satisfied are you with the current methods of training provided for new medical equipment?

Part 3: Innovation & Proposed Solution (Smart QR System)

9. Do you own or have access to a smartphone with a camera during your working hours?
10. If you could scan a QR code on a machine to instantly watch a digital training video, do you believe this would help you understand the equipment better?
11. How important is it to you that these training materials are provided in Kiswahili rather than English?
12. What specific features would make a digital training video most useful for you in your daily work?

13. What challenges or barriers do you think might prevent you from using a smartphone-based QR training system here at the hospital?

APPENDIX B: RAW DATA SUMMARY

Summary of data collected from face-to-face interviews with 30 healthcare workers at Kigamboni Hospital.

1. Demographics Summary (Responses to Q1-Q3):

Roles: We interviewed 15 Nurses (50%), 5 Medical Doctors (16.7%), 5 Laboratory Technicians (16.7%), and 5 Equipment Operators (16.7%).

Experience: 12 interviewees (40%) had 0–2 years of experience; 18 interviewees (60%) had 3 or more years of experience.

Departments: Interviewees were distributed across OPD (10), Maternity (8), Laboratory (7), and Surgery/Minor Theatre (5).

2. Challenges & Current Methods (Responses to Q4-Q8):

Confidence (Q4): The average confidence score was 2.1 out of 5. 65% of interviewees (19 out of 30) rated their confidence at a 2 or lower.

Difficulties Faced (Q5 & Q6): 80% (24 out of 30) admitted to facing difficulties. During the interviews, the most frequently described consequences were delayed patient care, fear of breaking expensive machines, and wasting time figuring out settings.

Current Training (Q7): 70% (21/30) stated they rely heavily on asking colleagues, 20% (6/30) rely on trial and error, and only 10% (3/30) attempt to read the printed English manuals.

Satisfaction (Q8): 83% (25/30) verbally expressed being "Dissatisfied" or "Very Dissatisfied" with current training methods.

3. Proposed Innovation Feedback (Responses to Q9-Q13):

Accessibility (Q9): 100% (30 out of 30) of interviewees confirmed they have access to a smartphone during their shifts.

Solution Acceptance (Q10): 93% (28 out of 30) agreed that scanning a QR code for a video tutorial would be highly beneficial to their workflow.

Language (Q11): 90% (27/30) stated that having the materials in Kiswahili was "Very Important."

Desired Features (Q12): Common themes expressed during the interviews included: videos must be short (under 3 minutes), clearly show the machine's buttons, and ideally be downloadable for offline viewing.

Potential Barriers (Q13): The most common concerns raised by interviewees were poor hospital internet/Wi-Fi connections.

9.1. REFERENCES

Ministry of Health Tanzania. (2023). Health Sector Strategic Plan.

World Health Organization (WHO). (2022). Medical Device Training Guidelines.

Tanzania Ministry of Health. (2024). Digital Health Strategy.

World Health Organization. (2021). Global Strategy on Digital Health.

United Nations Development Programmer (UNDP). (2023). Innovation for Sustainable Development.